One Stream, Many Windows: Soft Targets for Consolidating Fast and Slow Conversions

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Abstract. Conversion pipelines in advertising are split by speed: an upper-funnel onsite conversion is observable within minutes of the click, while a down-funnel offsite conversion takes a day or more. Each objective defines an *attribution window*, the elapsed time within which a conversion counts, and conventionally that window also fixes the *join delay*, the age at which the click's training record is written. Tying the two together forces every objective onto its own pipeline, each as stale as the window it waited for. We ask whether they can be set independently. Writing records sooner labels every conversion still in flight as a negative, and a streaming pipeline that trains each click at most once cannot revise them afterwards, yet most published corrections do exactly that: of 9 methods we reproduce, 7 write more than one record per click, up to 4.00. On the Criteo Delayed Feedback Dataset we compare three targets across 6,900 runs, five served capacities, three join delays, two attribution windows and two teacher architectures. Writing the record early and softening only its observed negatives with a teacher's estimate beats waiting for the matured label in 28 of 30 configurations, by 1.97% at seven days and 0.57% at one, which is 74% and 69% of what a label-peeking oracle attains. It is the strongest method obeying the one-record rule.

Keywords: Conversion rate prediction · Delayed feedback · Online advertising · Knowledge distillation · Streaming training pipelines.

1 Introduction

Advertising systems predict several kinds of conversion at once, and those kinds differ sharply in how long they take to observe. A view or an add-to-cart is recorded within minutes of the click; a purchase, an app install or a submitted lead may take a day or a week. Each objective is defined with an *attribution window*, the elapsed time within which a conversion is credited to the click, and a production system maintains several windows at once. Conventionally that window also determines the *join delay*, the age at which the click's training record is written to the stream: the pipeline waits until the label is knowable, then emits the record. The convention costs twice. A model for a slow objective trains only on data at least a window old, so it never sees recent behaviour. And a system that ranks several objectives jointly must either maintain one pipeline per objective or delay all of them until the slowest window closes, imposing the staleness of the slowest funnel on the fastest.

Neither cost is intrinsic. The attribution window is fixed by the business definition of a conversion; the join delay is a pipeline choice. We ask what happens when the two are set independently, and specifically whether an objective accountable to a seven-day window can be trained from records joined within 90 minutes.

The obstacle is that an early record is wrong in a known direction: every conversion still in flight when the record is written is recorded as a negative. A substantial literature corrects this censoring bias [1,14,13,2,4], but most corrections revise the record afterwards, by duplicating the click when its outcome lands, by relabelling it, or by adding prediction heads at further horizons. A streaming pipeline that appends each record once and trains on it once can do none of these. We state that restriction as a *single-visit* constraint and measure which published methods satisfy it. What then remains is a choice of what the single frozen target should contain, and the deficit must be filled at write time by an estimate from a model that has seen matured outcomes. Such a model is readily available in production, where large ranking models are already trained across objectives on fully matured data [10].

The contributions of this work are: (1) a separation of the join delay from the attribution window, with the causality and single-visit rules a deployable pipeline must satisfy, and an audit of 9 published methods against them, of which 7 violate single-visit by writing up to 4.00 records per click; (2) evidence that writing the record early and softening only its observed negatives beats waiting for the matured label, by more as the funnel slows, and is the strongest target obeying the single-visit rule; (3) the finding that at small served capacities the hard label can be dropped altogether, since a target consisting only of the teacher's estimate beats waiting in all 12 of 12 such cells, by 0.60% at one day and 0.85% at seven, which gives retrieval and other small-model stages a simpler option that carries no observed label at all.

2 Problem Setting and Proposed Approach

A click i occurs at time C_i with features x_i and, if it converts, does so after a delay $D_i > 0$, with $D_i = \infty$ when it never converts. The attribution window v belongs to the objective and fixes the target $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$. The join delay o belongs to the pipeline and fixes when the record is written, so the label available at that moment is

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

Setting $o = v$ reproduces the conventional per-objective pipeline. Setting $o \ll v$ is what we want, and the inequality in Eq. (1) is its price. We write $F(o) = P(D \le o \mid D \le v)$ for the fraction of the objective's conversions already visible when the record is written. Training proceeds in hourly rounds, each consuming the records released since the previous cut-off and then scoring every click of the following hour exactly once against the matured target $Y^{(v)}$. Every method

is therefore judged on the business label whatever it trained on, and updating before predicting keeps the evaluation causal.

Two rules a deployable pipeline must satisfy. **Rule 1, causality.** A record's label must be computable at the moment the record is written, and must not change afterwards. Assigning $Y_i^{(v)}$ to a record written at $C_i + o$ uses an outcome that has not yet occurred. Such an arm is an upper reference, not a system, and we report it under the name ORACLE. **Rule 2, single-visit.** Let n_i be the number of times click i enters the served model's loss over its lifetime. Rule 2 requires $\max_i n_i \leq 1$. Without it the question is empty, since any join delay can be shortened by writing a record early and writing it again later. We take the maximum over clicks rather than the mean, because it is the existence of a second write that obliges the pipeline to support revision, and the rule governs the served stream only: charging training-time auxiliaries would eliminate FSIW, ULC and our own arms alike.

Which published methods comply. We reproduced 9 published methods from scratch under a single protocol and counted the records each one's served model consumes per distinct click. 7 write more than one.[1] At $v = 1\,\mathrm{d}$ the counts are FTP [7] 4.00, MISS [8] 3.05, PI [15] 2.00, DDFM [3] 1.89, DFM [1] 1.33, DEFUSE [2] 1.03, ES-DFM [13] 1.03 . The highest, FTP, writes 4.00 records for every click it touches. The plain re-ingestion baseline [6,5] measures 1.03 and fails the same criterion. Duplication is a property of emission logic rather than of the model, and the audit returns identical counts at the smallest and largest served capacities. Only FSIW, ULC satisfy Rule 2. Of the two, FSIW corrects the loss and leaves the target itself untouched, while ULC changes the target, which makes it our closest comparator.

Four targets. What remains is a choice of what the single frozen target should contain. We study four answers, sharing a pipeline, schedule, feature set and optimiser, and differing only in the target $\tilde{y}_i$ and in the age Δ at which the record is written. FRESH writes at $\Delta = o$ with $\tilde{y}_i = Y_i^{(o)}$, filling the deficit with nothing. WAIT writes at $\Delta = v$ with $\tilde{y}_i = Y_i^{(v)}$, avoiding the deficit by deferring the write. It has a correct label on data of age v, and it is the conventional per-objective pipeline, so it is our baseline throughout. A practitioner who refuses both to duplicate records and to train on wrong labels is already running it. CORRECT-DISTILL writes at $\Delta = o$, keeps the observed label and softens only the observed negatives:

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \;\Big|\; Y^{(o)}{=}0,\, x\Big)\,. \tag{2}$$

[1] Two measurement traps both understate duplication. The denominator must be distinct clicks touched, since dividing records consumed by records released conflates duplication with variation in click volume. And a scoring window shorter than a method's revisit horizon cannot observe the revisit, so we score only clicks whose entire horizon lies inside the stream.

An observed positive stays exactly 1, and only the uncertain mass moves. This is the functional form used by ULC [11]; the two differ in how w^T is estimated, so comparing them isolates the estimator and nothing else. DISTILL-ONLY goes further and regresses on $\hat{p}_v(x) \approx P(Y^{(v)}{=}1 \mid x)$ alone, so the record carries features and a number but no observation at all. Finally, ORACLE writes at $\Delta = o$ but labels the record $Y_i^{(v)}$. It violates Rule 1 and is not deployable. We include it because it bounds what any causal correction could recover, which is what turns a positive result into a statement about remaining headroom.

The teacher. The estimates w^T and $\hat{p}_v$ come from a separate model trained on the *matured* stream, in which a record enters at $C_i + v$ carrying $Y_i^{(v)}$. Both quantities are read out from one shared trunk. The teacher is *evaluated* at the instant the student's record is written: at $C_i + o$ the student queries it and freezes the returned number into the record. Rule 1 is therefore satisfied even though the target is not binary. Softness is a different way of satisfying causality, not an exemption from it. The construction is distillation from privileged information [12], in which the teacher observes the matured label that the student by construction cannot. Being used only during training, the teacher costs nothing at serving time, though it does mean served capacity is matched across arms while total training capacity is not, so we treat teacher architecture as an experimental axis. We also distinguish two ways of maintaining it. A *continuously retrained* teacher keeps consuming the matured stream throughout the run, so deploying it means operating a second training job alongside the first. A *fixed* teacher is trained once during causal initialisation and never updated, so deploying it means shipping a static model file. The distinction is invisible in Eq. (2) and decisive operationally, so we measure both.

3 Experiments and Results

3.1 Experimental Setup

Data and protocol. We use the public Criteo conversion logs released with [1], the only large public advertising corpus carrying per-click conversion timestamps. Clicks span 60 days. Days $[0, 30)$ initialise the model causally and days $[30, 60)$ form the streaming period, divided into 719 hourly rounds yielding 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a seven-point grid on a held-out slice of the initialisation period, before any streaming evaluation, then pinned across seeds. The teacher's rate is fixed and decoupled from the student's, so the student's sweep tunes one model rather than two.

Model architecture. Every arm drives the same production-shaped backbone, so a method never sees a different model from its comparators. The eight integer fields enter a dense arm as $\log(1 + x)$ paired with an observed-mask bit, normalised and passed through a two-layer perceptron to a projection vector. The nine

categorical fields are hashed per field and embedded, one identifier per field, giving a token per field. Tokens and the dense projection are then consumed by a stack of Wukong interaction layers, each of which takes a token set and returns a token set of the same width so layers compose, combining a factorisation-machine block for explicit pairwise interaction with a linear compression block carried as a residual. The final token set is flattened to a single logit. Auxiliary networks that a baseline needs, such as a delay head or an importance-weight classifier, are held at a fixed size and never scaled with the served model, so the capacity axis measures the served model alone.

Scaling the served model. The five rungs M_0 to M_4 move three dials together rather than one: the interaction trunk is taken from successive rungs of a compute-efficient frontier, the hash width per categorical field grows $36 \to 256 \to 1{,}280 \to 16{,}384 \to 131{,}072$, and the embedding dimension widens $4 \to 4 \to 8 \to 8 \to 8$. Scaling only table rows against a fixed trunk would make the axis a feature-resolution axis rather than a capacity one, since on this corpus the trunk dominates the parameter count until the tables are large. The result is roughly an order of magnitude per step, 1,890 to 9,580,477 parameters and 0.001 to 0.286 MFLOPs per inference. Parameters and compute do not track each other: M_0 and M_1 differ 5.2-fold in parameters at identical compute because they share a trunk rung and differ only in table rows. We therefore report both rather than let one stand for the other.

Axes. Two attribution windows, $v = 1\,\mathrm{d}$ and $v = 7\,\mathrm{d}$, stand for a fast and a slow funnel; the contrast between them is the paper's point, so we never average across them. Three join delays are used, $o \in \{5, 30, 90\}$ minutes, crossed with the five served capacities. The fourth axis is the teacher, for which we study two architectures. *Teacher 1* places 8-dimensional embedding tables under a two-layer perceptron, giving 9,456,897 parameters at 0.039 MFLOPs. *Teacher 2* places 48-dimensional tables under a deep interaction trunk, giving 57,272,515 parameters at 1.593 MFLOPs. Running both is what lets us report the method's sensitivity to the choice of teacher rather than to one arbitrary configuration of it.

Scale and comparability. The evidence comprises 6,900 runs in two campaigns: 4,850 runs over 37 arms reproducing the published methods under teacher 1, and 2,050 runs over 14 arms adding teacher 2, with 10 seeds on the headline arms and 5 elsewhere. The second campaign runs a frozen copy of the first's harness and re-executes 850 of the first's runs, spanning 7 arms, under its own source revision. Every one returns the identical NE, Wait included across all 150 of its runs. That measured reproduction, rather than an assumption that two campaigns are comparable, is what licenses reading an arm from either campaign as a gain against Wait on one axis.

Metric and statistics. We report normalised entropy against $Y^{(v)}$, lower being better, pooled over all scored clicks rather than averaged over rounds, since a per-round average would normalise each hour by its own base rate and discard diurnal

variation a model can legitimately predict. WAIT itself attains NE between 0.72 and 0.81 across the grid, which is the base the relative figures below are read against. Results are the relative reduction against WAIT within a cell, computed per seed and then averaged, so a positive number means the short join delay is better than waiting for the matured label. Seeds differ only in initialisation and minibatch order, so seed i of two arms forms a matched pair: we use a paired t-test within each cell and Holm–Bonferroni correction across the 15 cells of a contrast, counting only cells whose gain is positive and survives. Seed dispersion is small away from FRESH, with median NE standard deviation 0.00107 and maximum 0.00520. FRESH reaches 0.0490 at $o = 5$ minutes, a finding rather than a nuisance: an uncorrected censored target at the shortest join delay is not a stable estimator. Seeds measure initialisation variance on one data origin, establishing precision rather than generalisation.

3.2 Findings

A short join delay beats waiting, and by more on slow funnels. Writing the record 2,016 times sooner than the seven-day window it answers to, and letting the teacher stand in for the conversions not yet seen, reduces NE in all 15 seven-day cells, of which 15 survive Holm–Bonferroni correction. The reduction ranges from 0.48% to 2.68%, with a mean of 1.97%. At the one-day window the same substitution gains 0.57% on average, positive in 13 of 15 cells and significant in 10. The two windows differ by a factor of three, and the direction is the informative part. The slower the funnel, the more a short join delay is worth. This inverts the natural expectation, since a longer window censors more of the label and leaves the teacher more to supply. What dominates instead is the cost of the alternative: WAIT at $v = 7\,\mathrm{d}$ trains on week-old data, and no amount of label accuracy compensates for that. Both ends of the achievable interval are measured. ORACLE, which writes at o but labels the record $Y^{(v)}$, gains 2.68% and 0.83%, so CORRECT-DISTILL recovers 74% and 69% of what any causal correction could attain. FRESH, the same early write with no correction at all, loses 20.1% and 11.8%. The deficit the teacher fills is an order of magnitude larger than the gains above it.

At small capacity the hard label can be dropped entirely. The result above uses the hybrid target of Eq. (2). The fully soft target of DISTILL-ONLY, which carries no observed label at all, shows how far the idea can be pushed. At the two smallest served capacities it beats WAIT in all 12 of 12 cells, by 0.60% at $v = 1\,\mathrm{d}$ and 0.85% at $v = 7\,\mathrm{d}$. That regime is the one retrieval and other early-funnel stages operate in, and the target suits it: the record reduces to features and a single number, emission needs no join against the observed outcome, and the pipeline still gains over waiting.

The target does not carry to the larger rungs. At $v = 1\,\mathrm{d}$ it loses to WAIT by 0.51% across the three largest capacities, crossing over between M_1 and M_2 at every join delay as Fig. 1 shows, and at $v = 7\,\mathrm{d}$ it stays ahead by only 0.10%. A purely soft target plausibly acts as a regulariser, supplying a variance reduction

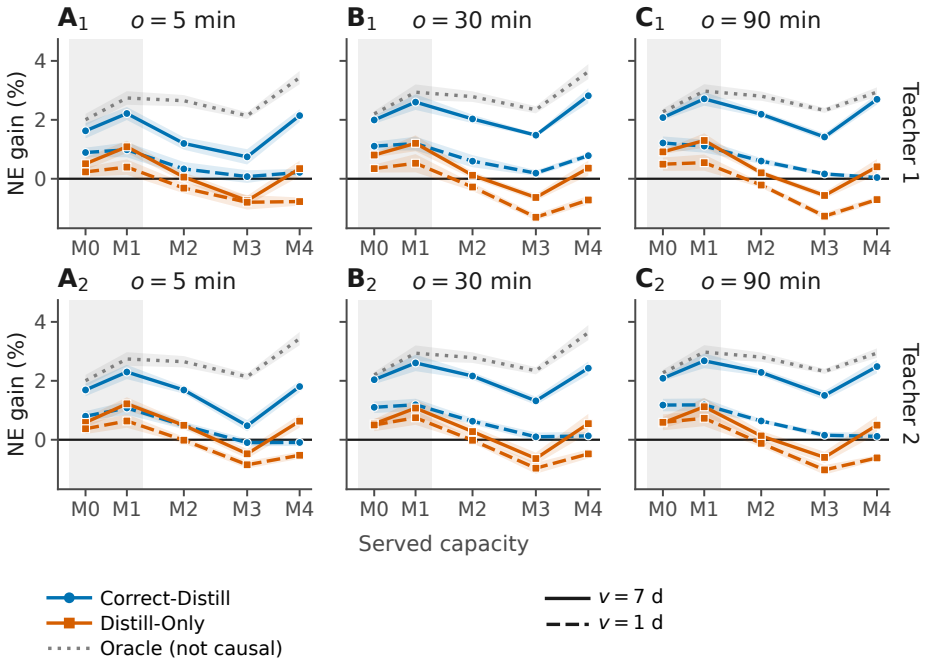


Fig. 1. Relative NE reduction against Wait as a function of served capacity. Columns are the join delay, (A) $o = 5$, (B) $o = 30$ and (C) $o = 90$ minutes; rows are the teacher the targets were distilled from, teacher 1 in A_1–C_1 and teacher 2 in A_2–C_2. Each row is read against the Wait arm of its own campaign, and the near-identical rows are the teacher-robustness result. Above the zero line beats the conventional per-objective pipeline. Colour denotes the target and line style the attribution window; Oracle, which labels an early record with the matured outcome and is therefore not causal, is drawn at $v = 7$ d to bound each panel. Shading on a curve is the standard deviation of the per-seed paired differences; the grey vertical band marks the two smallest capacities, the regime in which Distill-Only needs no hard label.

that a small model can use and a larger one no longer needs [9]. Where capacity is ample, retaining the observed positive is what pays: Correct-Distill beats Distill-Only in all 30 cells, and Wait in 28 of 30. The 2 exceptions both occur at $v = 1$ d and $o = 5$ minutes, at rungs M3, M4, the worse by 0.097%, and neither survives correction. Their location is coherent rather than random: the shortest join delay at the fast window is where the teacher has least to add, since 0.355 of the conversions are already visible.

Comparison with published methods at a production-like operating point. Table 1 fixes one cell rather than averaging the grid: the fast window at the longest join delay, $v = 1$ d and $o = 90$ minutes. That is the cell a large-scale advertising system sits in, where an upper-funnel objective closes within a day and the pipeline still needs the better part of an hour to join. It pools both campaigns

Table 1. All reproduced methods at $v = 1\,\mathrm{d}$ and $o = 90$ minutes, the operating point closest to a large-scale production setting, at the smallest and largest served capacities. NE and calibration error are absolute and lower is better; ΔNE is the per-seed paired reduction against WAIT in percent, higher is better, reported with the half-width of its 95% t interval. The replicate is a seed, five per arm and ten for WAIT, and a (teacher, seed) pair for the two arms in bold, which are ours and ran under both teachers. ‡ORACLE writes one record but labels it with the matured outcome, so it violates Rule 1 and is an upper reference rather than a method.

	M_0 (smallest)			M_4 (largest)		
Method	NE	Cal.	ΔNE (%)	NE	Cal.	ΔNE (%)
Single visited: yes						
WAIT	0.7865	0.0578	ref.	0.7209	0.0737	ref.
FSIW [14]	0.7785	0.0553	$+1.03 \pm 0.26$	0.7173	0.0781	$+0.51 \pm 0.09$
ORACLE‡	0.7769	0.0489	$+1.23 \pm 0.28$	0.7194	0.0617	$+0.22 \pm 0.08$
CORRECT-DISTILL	0.7772	0.0555	$+1.19 \pm 0.12$	0.7203	0.0676	$+0.09 \pm 0.06$
ULC [11]	0.7773	0.0529	$+1.18 \pm 0.22$	0.7207	0.0707	$+0.04 \pm 0.06$
DISTILL-ONLY	0.7821	0.0672	$+0.56 \pm 0.14$	0.7256	0.0782	-0.65 ± 0.06
FRESH	0.7929	0.1855	-0.80 ± 0.25	0.7345	0.1813	-1.88 ± 0.10
Single visited: no						
DDFM [3]	0.7902	0.0632	-0.46 ± 0.31	0.7168	0.0572	$+0.57 \pm 0.09$
PI [15]	0.7806	0.0692	$+0.76 \pm 0.43$	0.7181	0.0811	$+0.40 \pm 0.03$
FTP [7]	0.7786	0.0515	$+1.02 \pm 0.18$	0.7187	0.0636	$+0.31 \pm 0.16$
DEFUSE [2]	0.7789	0.0681	$+0.98 \pm 0.30$	0.7217	0.0856	-0.10 ± 0.07
ES-DFM [13]	0.7791	0.0690	$+0.96 \pm 0.31$	0.7219	0.0869	-0.13 ± 0.08
DFM [1]	0.7569	0.0612	$+3.78 \pm 0.19$	0.7236	0.0584	-0.37 ± 0.09
MISS [8]	0.7830	0.0826	$+0.45 \pm 0.24$	0.7268	0.1229	-0.82 ± 0.23

and shows the two ends of the capacity ladder separately, because averaging them reports neither.

Four things follow. First, the two rungs disagree, which is why both are shown. At M_0 correcting the early write is worth $+1.19 \pm 0.12\%$ and very nearly exhausts what any causal correction could take, since ORACLE reaches $+1.23\%$ and we recover 97% of it. At M_4 everything compresses: ORACLE itself is only $+0.22\%$, and our $+0.09 \pm 0.06\%$ is 42% of that. This cell simply has little delay left to correct, because 0.803 of its conversions are already visible when the record is written. Second, the single-visit rule is cheap at small capacity and not at large: DFM leads at M_0 with $+3.78\%$, far above the single-visit ORACLE, because duplicating a click supplies training signal that matters most when the model has least capacity to extract it. Third, among methods a streaming pipeline can run we lead at M_0, where ULC [11] reaches $+1.18\%$ within our interval, but not at M_4, where FSIW [14] reaches $+0.51 \pm 0.09\%$ by correcting the loss rather than the target. FSIW averages -0.45% across this window's 15 cells, so that rung flatters it rather than describing it, and across the slow window's 15 cells we lead ULC by $+0.13\,\mathrm{pp}$ in 12 of 15 cells. Fourth, the gain does not cost calibration.

CORRECT-DISTILL is better calibrated than WAIT at both rungs, while FRESH, the same early write left uncorrected, degrades calibration error by 3.2 and 2.5 times. Softening the observed negatives repairs the censored target's bias, not merely its ranking.

Robustness to the teacher, and what bounds it on this corpus. Substituting teacher 1 for teacher 2 changes little. CORRECT-DISTILL gains 2.00% with teacher 1 against 1.97% with teacher 2 at seven days, and 0.63% against 0.57% at one; over all 30 cells the two architectures differ by -0.043 pp for CORRECT-DISTILL and +0.114 pp for DISTILL-ONLY. Holding parameters fixed while spending 4.7 times more teacher compute stays inside the same margin. The teacher's learning rate is verified rather than inherited: the default beats a rate ten times smaller in every cell tested and matches one twice as large, so the insensitivity describes a converged teacher rather than an untuned one. The practical reading is that the method does not depend on a particular teacher, which is what makes it portable across pipelines.

That insensitivity is a property of the corpus as much as of the method. Criteo carries eight integer and nine hashed categorical fields with no creative, content or user-history signal, so $P(Y^{(v)}{=}1 \mid Y^{(o)}{=}0, x)$ is close to exhausted by a modest architecture and further teacher capacity has little left to condition on. A production corrector would see far richer features and longer histories, and a cross-objective foundation model trained on fully matured outcomes [10] is precisely the artefact positioned to supply it. We expect such a teacher to sharpen the corrector and widen the gains reported here, but that is an expectation about a setting we did not measure, and we claim only the Criteo result.

Finally, fixing the teacher after causal initialisation, so that it is shipped as a static model rather than maintained as a second training job, costs 0.08 and 0.27 pp, and the fixed-teacher arm still beats WAIT in 27 of 30 cells. Adoption therefore does not require standing up a continuously retrained auxiliary alongside the served model.

4 Concluding Remarks

An objective's attribution window is fixed by the business definition of a conversion, but the join delay its pipeline adopts is not, and the two can be set independently. Shortening the join delay by three orders of magnitude improves accuracy rather than degrading it, recovers 74% of what a label-peeking oracle attains at seven days, and helps most where the funnel is slowest. Which target to write depends on the capacity being served: above the two smallest rungs the record should retain its observed positive, while at small capacity the label can be dropped and the teacher's estimate carried alone, which suits retrieval and other early-funnel stages and is simpler to emit.

The immediate consequence is for training several ranking objectives in one model, an active industrial direction [10] whose first obstacle is a pipeline obstacle rather than a modelling one. Under WAIT, objective k's record exists only at

$C_i + v_k$, so a shared append-only stream must either write K records per click, violating Rule 2 K times over, or delay every objective until $\max_k v_k$, imposing the slowest funnel's staleness on the fastest. A single stream at one short join delay instead carries $\tilde{y}_{i,k} = Y_{i,k}^{(o)} + (1 - Y_{i,k}^{(o)})\, w_k^T(x_i)$ for every objective at once, because w_k^T is a function the pipeline evaluates rather than an outcome it awaits. ULC reaches comparable accuracy, so the choice between them is architectural: ULC fits each w_k inside the served model's training loop, turning K objectives into K auxiliary heads on one optimiser, whereas a teacher places each w_k upstream with a single number written onto a record as its only interface. Section 3 shows the result is stable across teacher architectures and survives freezing the teacher after initialisation, so consolidation can add K static model files to the build rather than K running training jobs.

Three limits bound the claim. We ran one objective at a time, so we have no evidence on interference between objectives sharing a trunk, which is the central risk in the multi-task target above. Our teacher stands in for the production ranking model the argument invokes rather than being an instance of one. And one corpus with one chronological split measures precision rather than generalisation. The experiment this makes worth running is K prediction heads, K attribution windows, one stream, and the teachers placed upstream.

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